Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/17/2013
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Microgravity and Magnetic Fields - Directional Root Growth and Gene Expression		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/03/2010	**End Date:**	03/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: New end date is 3/31/2015 per NSSC information (Ed., 1/22/14) NOTE: New end date is 12/31/13 per NSSC information (Ed., 8/6/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10AP91G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal combines morphological measurements on micro-gravity germinated and grown (Brassica) roots that are exposed to high-gradient magnetic fields (HGMF) with a high-resolution spatial analysis of genes involved in gravitropism. Magnetic gradients exert a repulsive force on diamagnetic substances such as the starch in amyloplasts, which results in gravitropic-like curvature. The proposal is based on hardware that was developed for a previous flight experiment on STS-107. As in the originally proposed experiment, microscopic analysis of amyloplast size and distribution will be studied in the context of achieved curvature. In addition, high-resolution genetic profiling will be conducted that focuses on the cytoskeleton and starch development. The result of the proposed work will enable directional growth of seedlings and eventually mature plants under microgravity conditions using HGMF. Three conditions will be used to differentiate microgravity from uniform magnetic fields and HGMF conditions. Only HGMF is expected to influence root curvature. A uniform magnetic field will serve as control for possible secondary magnetic effects. This condition will be compared to an in-flight control without magnetic field to determine growth rate, gene expression, growth, and directionality (autotropism). Root material from these experiments will be used for investigations on cytoskeletal organization and the expression of starch-related genes. A second aspect of the proposed work will investigate the process of gravisensing, whether dependent on amyloplast position, movement, or force exerted onto amyloplasts. Space experimentation is necessary to establish a threshold value for a directional magnetic force. In addition to providing insight in the fundamental mechanism of the gravisensing system of plants, the data from this experiment can be used to develop technology to provide a directional, gravity-like stimulus to plants during early stages of germination. The proposed experiment can be performed with existing hardware, be adapted easily to changing flight schedules, and requires minimal crew time. The anticipated benefits include a better understanding of root development, effects of amyloplast displacement on root development and curvature, and compensatory measures to alleviate microgravity conditions. A newly developed gene extraction technology is ideally suited for gene-profiling studies and has far-reaching implications for studying gene expression, especially for confined space flight experiments. Additional benefits of the project include training of students at the undergraduate and graduate levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to an improved understanding of basic biological questions, such as identification of the site of graviperception, curvature-induced (but not gravity-dependent) gene expression, the experiment also tests the effect of magnetic fields on plant roots. Therefore, biological questions are investigated in conjunction with physical principles. We intent to analyze the effect of static magnetic fields on biological systems. This work will provide a starting point for the current discussion of the detrimental effects of (electro)magnetic fields. In addition, understanding basic biological questions will improve the prospect for enhanced plant productivity, resource optimization, and reduced maintenance.
Task Progress:	The last year of the funding period continued to focus on the biology of Brassica seeds and seedlings and tests, laboratory and procedural changes at KSC. Preparation to payload verification and gene expression and growth studies were performed in the PI's lab. KSC continued with the development of hardware, most notably the camera system, hardware integration, and MFC performance. The most important obstacle from last year's science verification test was the lack of seed germination. An extensive test of all involved parameters suggested that unknown volatile organic compounds (VOC's) were the reason for germination failure. Corrective measures included a preventative enclosure of activated carbon cloth (ACC), and changed washing and drying procedures. Tests after the implementation of these changes in triple sealed Generic Containment Unit (GCU) resulted in much improved germination to an extent that now the removal of ACC is being examined. Recent advanced in the hardware development were complicated by a major systems failure during vibration tests. The experimental load exceeded the target load and resulted in breakage of the base plate, the loosening of the computer stack, failed MEDUSA motor, and leakage of MFC's. These problems are being corrected in time for a scheduled PVT in October 2013. Activities in the PI lab focused on the storability of seedlings in RNALater, the longevity of formaldehyde-based fixative in the presence of rust derived from corroding magnets, immuno-staining procedures, and the detectability of gene expression changes associated with germination, gravitropic curvature, and growth. The effect of rust on fixation and staining was examined and found to be negligible when proper measures were implemented. These steps include washing of fixed material, frequent exchanges of buffer before sectioning and staining, and slightly modified staining protocols. RNA studies showed that custom-made RNALater fixative induces osmotic responses that lead to the degradation of some of the RNA. The continuation of solid phase gene extraction led to a precise description of temporal and spatial changes in gene expression in roots corresponding to those that will be used for the flight experiment. A paper describing these relationships has been submitted for publication. Original studies with a Labchip 90 system have now been continued with a different but related system (Qiaxcel). We were able to characterize the quality of mRNA that was extracted from young (< 5d old) roots.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Articles in Peer-reviewed Journals	Hasenstein KH, Scherp P. "Biotechnology in space: Challenges and opportunities for solid phase gene extraction." Current Biotechnology. 2013;2(3):175-8. (originally reported as "in press, as of August 2013.") http://dx.doi.org/10.2174/22115501113029990014 , Sep-2013
Articles in Peer-reviewed Journals	Hasenstein KH, John S, Scherp P, Povinelli D, Mopper S. "Analysis of magnetic gradients to study gravitropism." Am J Bot. 2013 Jan;100(1):249-55. Epub 2012 Nov 21. http://dx.doi.org/10.3732/ajb.1200304 ; PubMed PMID: 23174915 , Jan-2013